MESSENGER STUDIOS · VIDEO TEAM

Permanent Stage Build Manual

A step-by-step guide to building the 15-panel modular production stage — written for builders with no carpentry experience.

Messenger International · Spring Hill, TN
Rev 0 · June 2026 · All measurements verified

Messenger Studios — Permanent Stage · Build Manual

A step-by-step guide for building the 15-panel modular production stage that sits in front of the curved LED wall.

Written so that anyone on the Video Team can build it — no prior carpentry experience assumed. Every measurement here has been double-checked against the stage blueprint. Build it **fastener-only** (screws, bolts, nails, dowels) — there is no glue in this build, which keeps every panel removable and serviceable.

> **Read this whole page once before you cut anything.** Then work part-by-part. The hard part is the 5 curved panels (Part 7) — everything else is repetitive and easy.

What you are building

A **20-foot-wide stage**, **7 inches tall**, made of **15 separate panels** that bolt together. Ten panels are plain 4'×4' squares. Five panels (the back row, against the LED wall) have a curved back edge so the stage hugs the wall's curve.

Spec	Value
Overall width	**20'-0"** (5 panels across)
Depth at center	**12'-0"** (3 panels deep)
Depth at the wings (sides)	**9'-0"** (curved back edge)
Height	**7"** (level), adjustable 7"–8" with the leveler feet
Panels	**15 total** — 10 square, 5 curved
Panel size	**4'×4'** (48"×48") each

Each panel is a wooden box (2x6 frame) with a plywood top, a thin black Masonite skin, four leveler feet, and a 6"-wide cable tunnel running front-to-back through its middle. When all 15 are bolted together, the tunnels line up into 5 channels that let you run cables anywhere under the stage and reach them through pop-out doors in the front skirt.

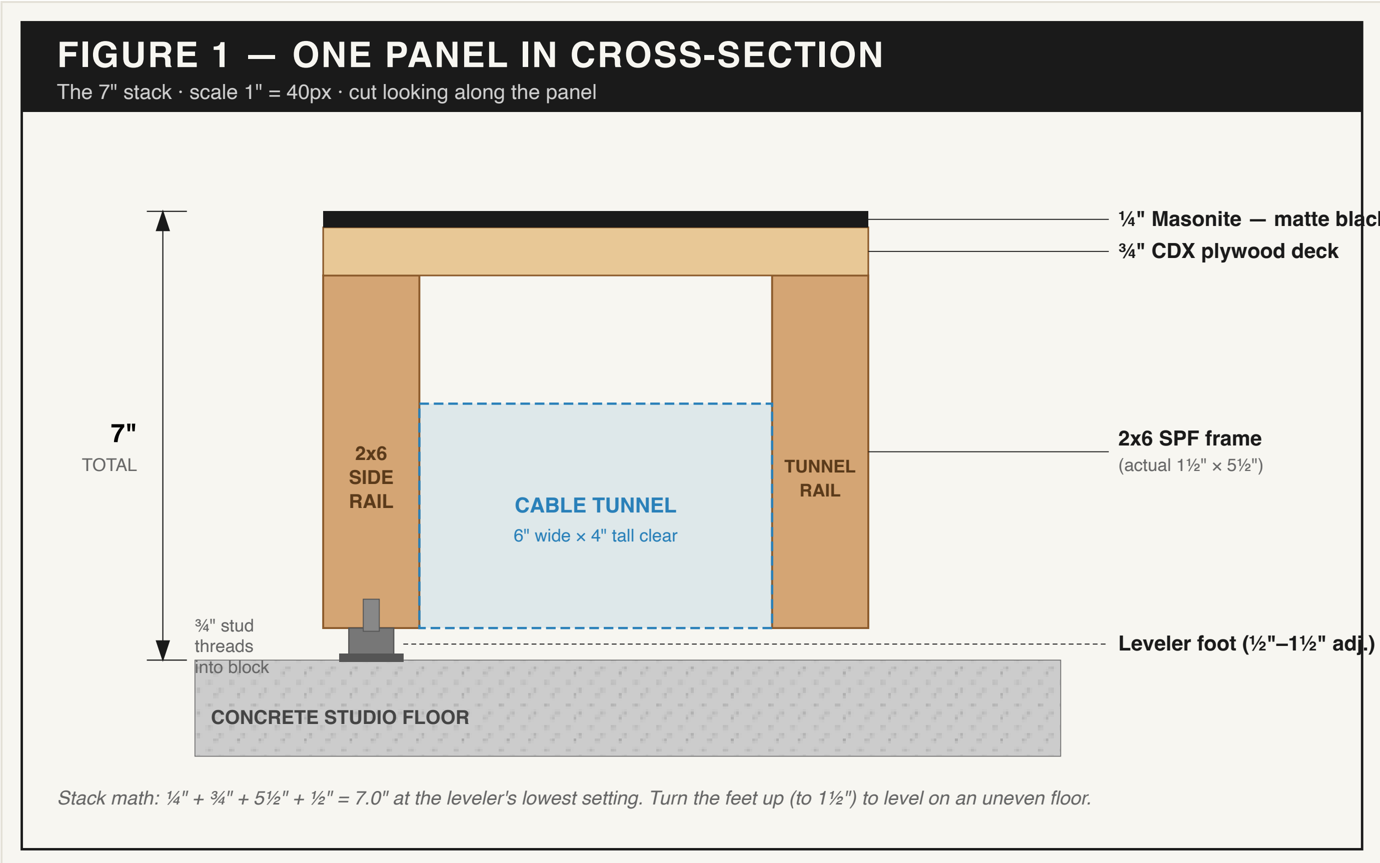

Figure 1 — Cross-section of one panel. Bottom to top: leveler foot, 2x6 frame, ¾" plywood deck, ¼" Masonite top = 7" total.

⚠️ Safety first (read before every work session)

- **Eye protection, ear protection, and a dust mask** every time a saw runs. Masonite and MDF dust is fine and nasty — wear the mask.
- **Two people** to lift any finished panel (each weighs ~60–80 lb).
- **Clamp your work** before sawing. Never cut a board you're holding by hand.
- **Keep fingers 6" from any blade.** Let the saw stop fully before setting it down.
- **Mind the cords** — keep them behind the saw, never across the cut line.
- **Tidy as you go.** Offcuts and screws on the floor are trip and puncture hazards.

Tools you'll need

You are **not** buying these — borrow, rent, or use what the team has. (These are tools, not part of the stage materials.)

Cutting

- ☐ Circular saw **or** table saw (straight cuts)
- ☐ **Jigsaw** (curved cuts — required for the 5 back panels)
- ☐ Miter saw *(optional, makes square cuts faster)*

Drilling / driving

- ☐ 2× cordless drill/driver (one for pilot holes, one for screws — saves bit swapping)
- ☐ Impact driver *(optional, easier on screws)*
- ☐ Bit set: countersink bit, ⅜" spade or auger bit (bolt holes), ¼" brad-point (dowel holes), pilot bits, Phillips/square driver bits
- ☐ Adjustable wrench or ⅜" socket (for the wing nuts)

Measuring / marking

- ☐ 25' tape measure
- ☐ Framing square **and** speed square
- ☐ Chalk line or a long straightedge (4'+)
- ☐ Pencil + fine marker
- ☐ A **flexible batten** — a thin, springy strip about ¼" × 1" × 50". Rip one from a plywood or Masonite offcut. *(This is how you draw the curve — see Part 7.)*

Other

- ☐ 4+ bar clamps
- ☐ 2 sawhorses or a sturdy work table
- ☐ Sandpaper (80 & 120 grit) to ease cut edges
- ☐ Paint roller, 2" brush, tray (for Part 8)
- ☐ Wire strippers + small screwdriver (for Part 11, the LED strip)

Materials checklist (the whole build)

Everything below is from the original plan. Confirm you have it all **before** you start. Counts are the verified build totals.

Lumber

- ☐ **2x6 × 8' SPF, kiln-dried — 56 boards** *(the plan estimated ~50; the verified cut list with leveler blocks + curved-panel cleats lands at ~54, so buy 56 to avoid running short)*

Sheet goods

- ☐ ¾" CDX plywood, 4'×8' — **8 sheets** (becomes 16 deck squares; you need 15)
- ☐ ¼" tempered Masonite, 4'×8' — **8 sheets** (16 top skins; you need 15)
- ☐ ½" MDF, 4'×8' — **3 sheets** (front skirt + side skirts)

Fasteners & hardware

- ☐ #8 × 3" deck screws — **~1000** (frames + decking)
- ☐ #8 × 1¼" screws — **~500** (skirt, leveler blocks, magnet catches)
- ☐ #6 × ⅝" finish nails — **1 box** (Masonite top skin)
- ☐ ⅜" × 4" carriage bolts + ⅜" wing nuts + ⅜" washers — **~90 sets used** (plan supplies ~180 — keep the rest as spares)
- ☐ ¼" × 2" dowel pins — **~44 used** (plan supplies 60)
- ☐ Stage leveler feet, ¾" stud, ½"–1½" adjust — **60** (4 per panel)
- ☐ Neodymium magnet catches, 5 lb — **10** (2 per skirt door)

Finish

- ☐ Matte black stage/acrylic paint — **Behr Ultra Pure Black** (1–2 gal)

LED strip system

- ☐ RGB+W LED strip, 24V, IP65, 60 LED/m — **7 m**
- ☐ Aluminum channel + frosted diffuser — **7 m**
- ☐ 24V 200W power supply (Meanwell or equiv) — **1**
- ☐ RGBW DMX decoder (≥4-channel) — **1**
- ☐ 5-pin DMX cable to the lighting board — **1**

> ✅ **Why no glue?** The original plan has none, and you don't need it. Screws and bolts hold a stage like this rock-solid, and skipping glue means any panel can be unbolted and pulled out later for repairs or re-cabling. Just keep the screws snug.

Understand the layout

The stage is a grid: **5 columns** (A–E, left to right as the audience sees it) × **3 rows** (1 = front/downstage, 2 = middle, 3 = back/upstage against the wall). That gives 15 panels, named **A1** through **E3**.

- **Rows 1 & 2** (10 panels) are plain 4'×4' squares — all identical.
- **Row 3** (5 panels: A3, B3, C3, D3, E3) is curved on the back edge.
- The cable tunnel runs **front-to-back** down the middle of each column (5 tunnels total).

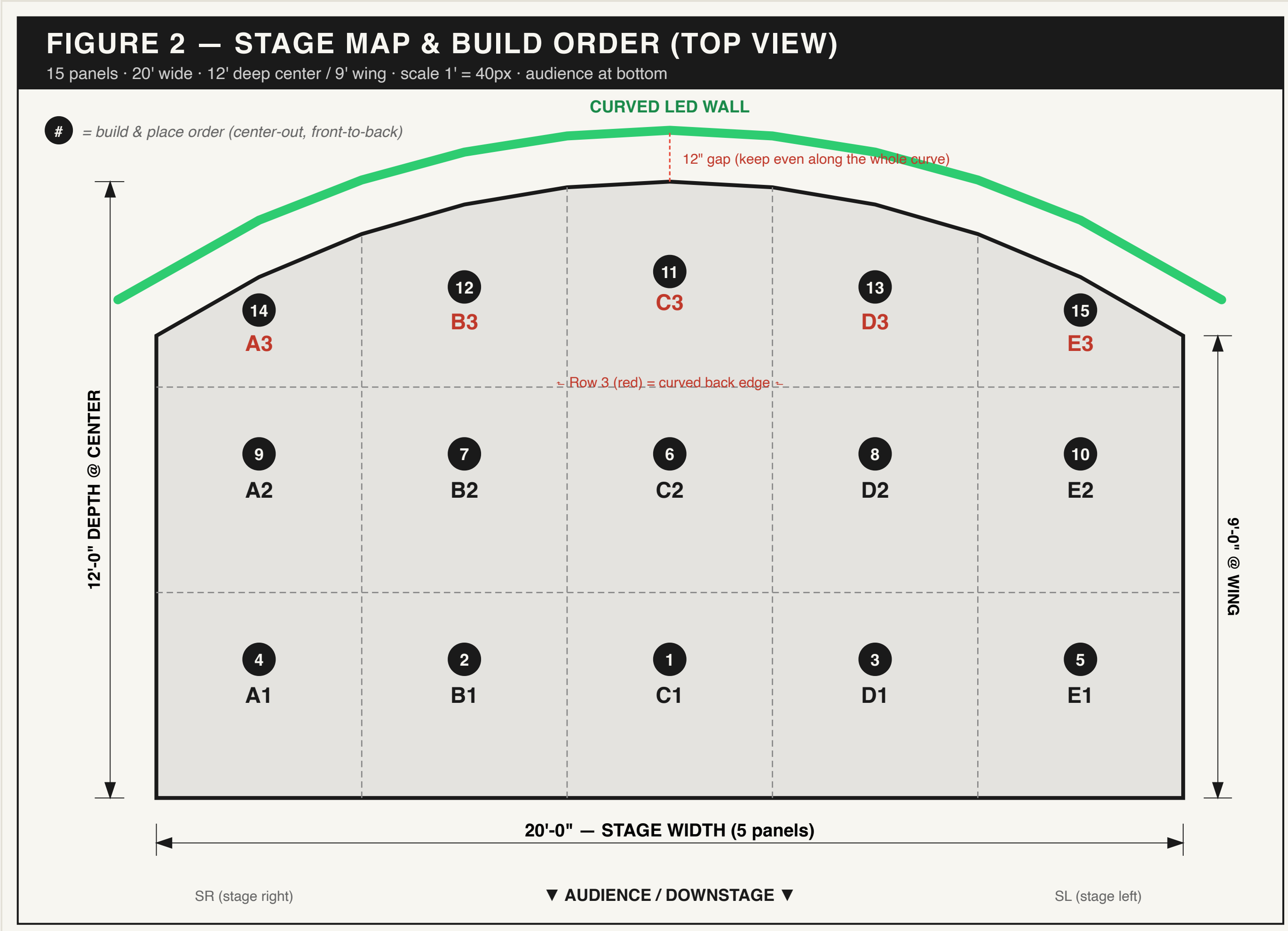

Figure 2 — Top-down map. Build and place panels in the numbered order. The back row (3) follows the LED-wall arc.

Orientation words used in this manual:

- **Downstage** = toward the audience (front of stage).
- **Upstage** = toward the LED wall (back of stage).
- **SR / SL** = stage right / stage left (the performer's right/left).

Cut list (cut everything first)

Cutting all your pieces up front ("batching") is faster and more accurate than cutting one panel at a time. Label every piece with a pencil as you cut.

Lumber — per panel

Every panel (square **and** curved) starts from the **same** frame. You'll cut the curve later, after assembly, on the 5 back panels only.

For ONE panel frame (×15):

Piece	Length	Qty	Notes
Front & back rail	**48"**	2	Each gets a notch — see below
Side rail	**45"**	2	Fits *between* the front & back rails
Tunnel rail	**45"**	2	Runs front-to-back, forms the cable tunnel
Mid-bay blocking	**18"**	2	Stiffens the frame
Leveler corner block	**5½"**	4	Solid wood for the leveler feet to thread into

> **The notch:** On each **48" front/back rail**, cut a notch in the **bottom edge**, **centered**, **6" wide × 4" deep (tall)**. This leaves a 1½" strip of wood above the notch (the deck still rests on it). When panels line up, these notches form the continuous cable tunnel. Cut notches with the jigsaw.

Cut math check: 15 frames × (2×48 + 4×45 + 2×18) = 15 × 312" = 4,680" of rail, plus 60 corner blocks (5½") + 10 curved-panel cleats (~24") ≈ **5,160" ≈ 54 boards.** Buy **56**.

Sheet goods — cuts

Material	Cut into	Yield	Use
¾" CDX plywood (8 sheets)	**48" × 48" squares**	16 (need 15)	Panel decks — one per panel
¼" Masonite (8 sheets)	**48" × 48" squares**	16 (need 15)	Top skins — one per panel
½" MDF (3 sheets)	**7"-tall strips**	plenty	Front skirt (5 × 48") + side skirts

> Cut deck and skin squares now, but **don't cut the curve into the back-row decks/skins yet** — you'll do that in Part 7, after the frame is curved, so the curve matches exactly.

Part 5 — Build a square panel (Rows 1 & 2)

Do **one** panel completely as a test, check it, then batch the other nine. Each finished panel = frame → tunnel → deck → skin → levelers.

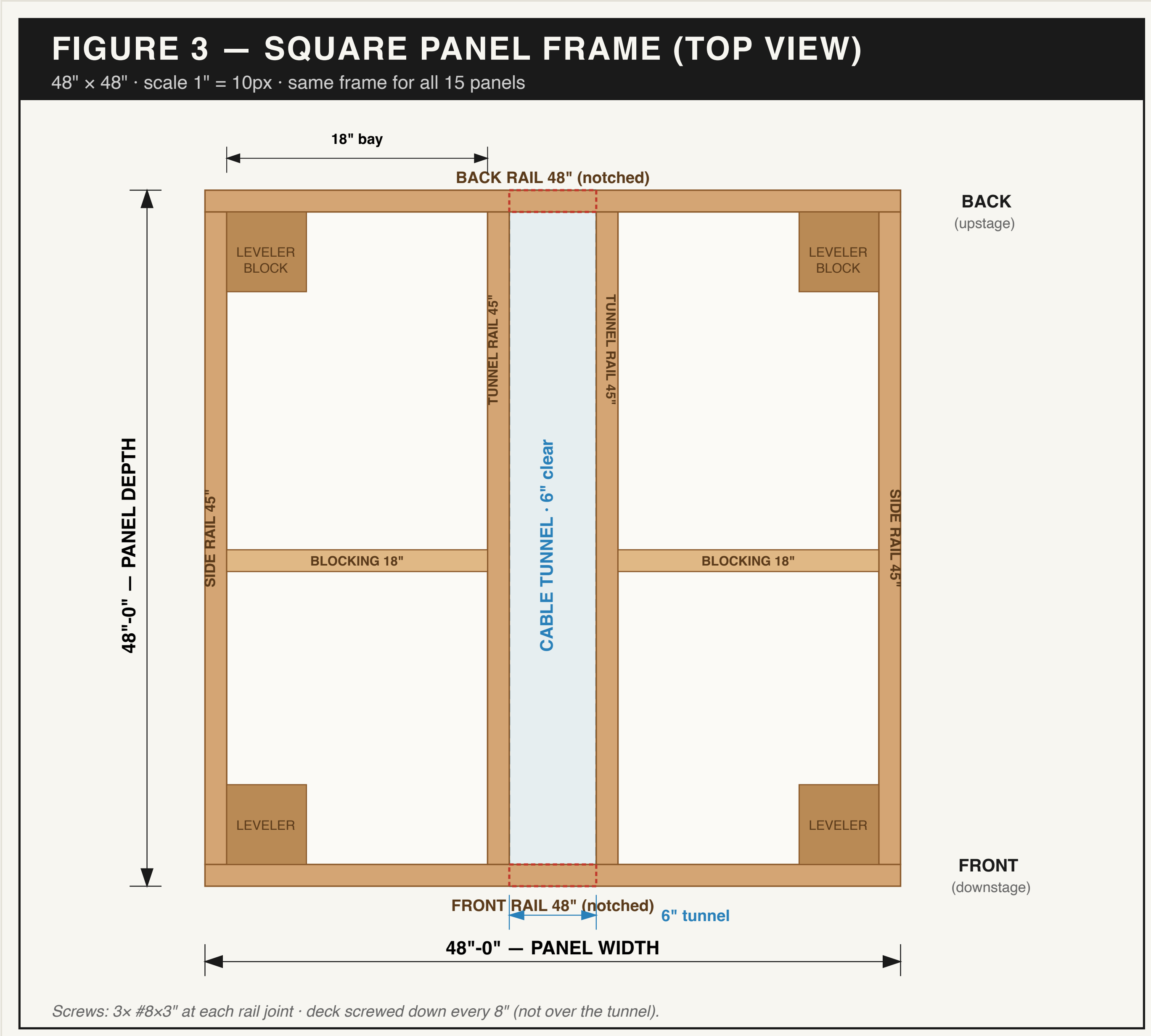

Figure 3 — Square panel frame, seen from above. All inside measurements are clear-gap dimensions.

Step 5.1 — Build the frame box

1. Lay the two **48" front/back rails** parallel, notch-side down, 48" apart (outside-to-outside = 48").
2. Set the two **45" side rails** *between* them, flush to the ends, forming a 48"×48" square box.
3. At each corner, drive **3× #8 × 3" screws** through the 48" rail into the end of the 45" rail. Pre-drill near the ends so the wood doesn't split.
4. Check the box is square: measure both diagonals corner-to-corner. **They must be equal** (≈67⅞"). Adjust until they match, then it's square.

Step 5.2 — Add the tunnel rails

1. The two **45" tunnel rails** run the same direction as the side rails (front-to-back).

2. Position them so there is a **6" clear gap between them, centered** in the panel. (Measure: the gap's edges sit at **21"** and **27"** from one side.) This 6" gap lines up with the notches you cut — it's the cable tunnel.
3. Screw each tunnel rail to the front and back rails: **3× #8 × 3"** at each end.

Step 5.3 — Add blocking

1. Drop one **18" blocking** piece into each side bay (the space between a side rail and a tunnel rail), **halfway back** (22½" from the front).
2. Screw through the side rail and the tunnel rail into each end: **2× #8 × 3"** per end.

Step 5.4 — Lay the deck

1. Set a **48"×48" × ¾" plywood** square on top. It should sit flush to all outside edges.
2. Screw it down with **#8 × 3"** screws, **countersunk flush**, every **8"** around the perimeter and along the tunnel rails and blocking. *(Flush heads matter — the Masonite skin must lie flat on top.)*
3. **Do not** put screws over the open 6" tunnel.

Step 5.5 — Add the Masonite top skin

1. Lay a **48"×48" × ¼" Masonite** square, smooth side **up**, on the deck.
2. Fasten with **#6 × ⅝" finish nails**: every **6"** around the edges, every **12"** in the field. Set the heads just below the surface.

Step 5.6 — Mount the leveler feet

1. Flip the panel upside down.
2. At each corner, screw a **5½" corner block** flat into the inside corner with **2–3× #8 × 1¼"** screws. This gives solid wood for the foot.
3. Install a **leveler foot** in each corner block per the foot's instructions (most thread into a pre-drilled ¾" hole or a small mounting plate that comes with the foot). Start them all at their **lowest** setting (~½").
4. Flip the panel back over. **One square panel done.** Repeat for all 10 (A1–E1, A2–E2).

> **QC each panel before moving on:** diagonals equal (square), deck flush, no proud screw heads, tunnel clear, all 4 feet thread smoothly.

Part 6 — Build the curved panels (Row 3) — the careful part

The 5 back panels (A3–E3) are built **exactly like a square panel first**, then the back edge is **marked with measurements and cut to the curve**. Take your time here.

> **Why measurements instead of a compass?** The curve's radius is **218 inches** (18 feet). You can't swing an 18-foot compass indoors. Instead you mark a row of dots at known depths and bend a flexible batten through them — this draws a perfect, accurate arc. This is the standard way pros loft a large curve.

Step 6.1 — Build 5 full square frames + decks

Build A3, B3, C3, D3, E3 as **complete square panels** through Step 5.4 (frame + tunnel + plywood deck). **Stop before the Masonite skin.** Leave them as 48"×48" squares for now.

Step 6.2 — Mark the curve on each back panel

Each panel gets a **different** curve. Here's how to mark one:

1. On the **plywood deck**, pick the edge that will face the **LED wall** (upstage). Along the **side edges**, measure **from the downstage edge** and make a tick at the depth given in the table. These depths are measured at **6" intervals** across the 48" width.
2. **u** = distance across the panel (0" at one labeled corner → 48" at the other). **Depth** = how far back to mark, measured from the front (downstage) edge of that panel.
3. Put a finishing nail lightly at each dot, **bend the flexible batten** so it touches every dot, and trace the curve in pencil along the batten.

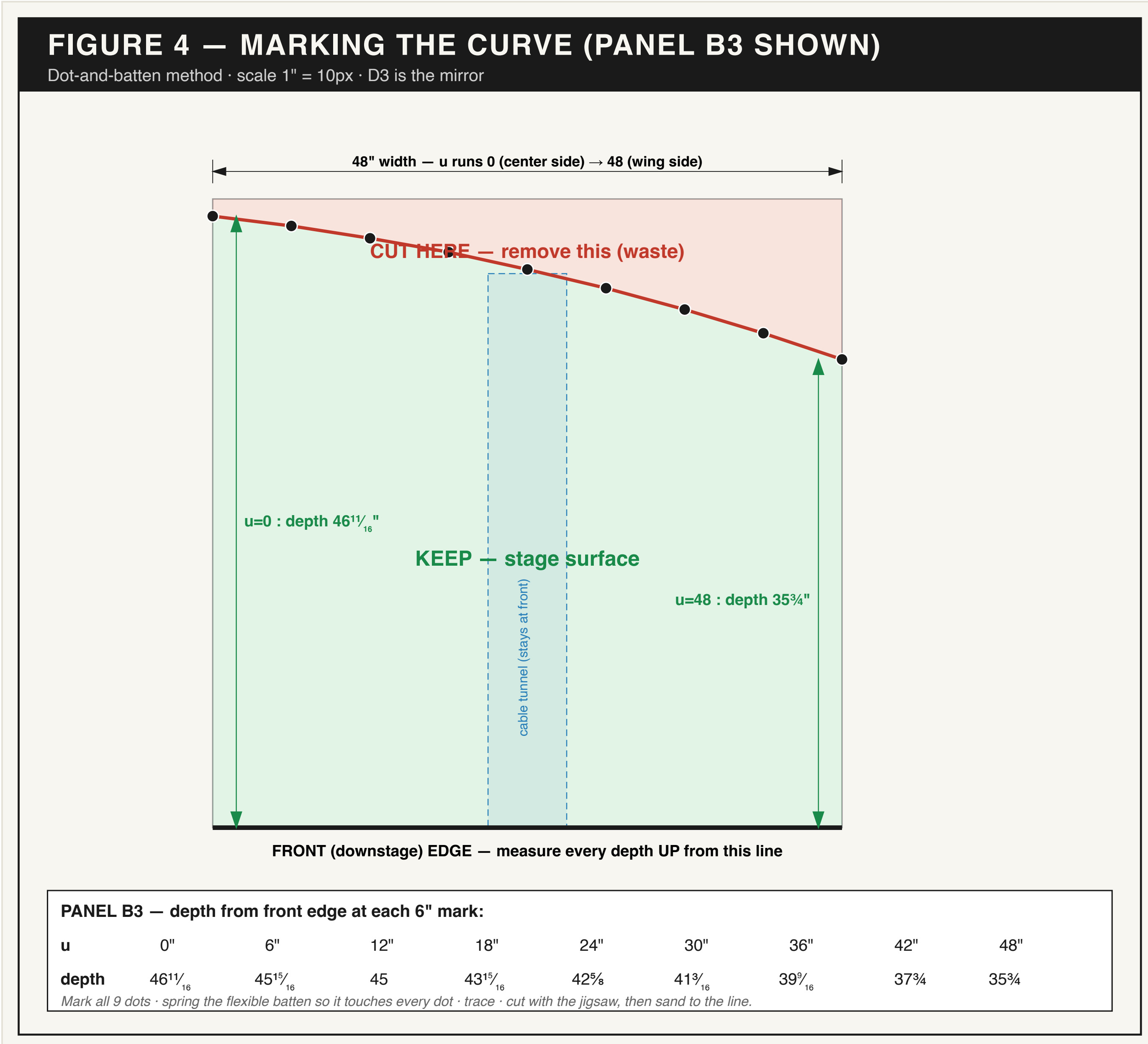

Figure 4 — The ordinate (dot-and-batten) method, shown on panel B3. Mark each depth, spring the batten through the dots, trace, cut.

Panel C3 (center) — barely curved:

u (across)	0"	6"	12"	18"	24"	30"	36"	42"	48"
Depth	$46\tfrac{11}{16}$"	47¼"	$47\tfrac{11}{16}$"	$47\tfrac{15}{16}$"	48"	$47\tfrac{15}{16}$"	$47\tfrac{11}{16}$"	47¼"	$46\tfrac{11}{16}$"

Panels B3 & D3 — D3 is the mirror of B3. Put **u = 0 at the edge facing the center** of the stage:

u (across)	0"	6"	12"	18"	24"	30"	36"	42"	48"
Depth	$46\tfrac{11}{16}$"	$45\tfrac{15}{16}$"	45"	$43\tfrac{15}{16}$"	42⅝"	$41\tfrac{3}{16}$"	$39\tfrac{9}{16}$"	37¾"	35¾"

Panels A3 & E3 (wings) — E3 is the mirror of A3. Put **u = 0 at the edge facing the center** of the stage:

u (across)	0"	6"	12"	18"	24"	30"	36"	42"	48"
Depth	35¾"	33 9/16"	31 3/16"	28 9/16"	25¾"	22 11/16"	19⅜"	15 13/16"	12"

Sanity check before cutting: Lay B3 next to C3 the way they'll sit on stage. The depth where they meet should match — C3's edge reads **46 11/16"** and B3's inner edge reads **46 11/16"**. Same for B3↔A3 at **35¾"**. If neighboring edges match, your curve is continuous. ✅

Step 6.3 — Cut the curve

1. Clamp the panel firmly, deck up.
2. With the **jigsaw**, cut along the traced line through the plywood **and** the 2x6 frame beneath it in one pass. Go slow on the thick frame parts. Stay just outside the line, then sand to the line.
3. Save the offcuts — you'll use straight bits for the cleats.

Step 6.4 — Add a back cleat

Cutting the curve removes most of the back rail, so add support along the new edge:

1. From straight 2x6 offcuts, cut **1–2 short cleats (~24" each)** and fit them just inside the curved cut, spanning between the cut ends of the side and tunnel rails.
2. Screw them to the deck from above (**#8 × 3"**, countersunk) and to any rail they touch. This stiffens the curved edge. *(This edge faces the wall in the 12" gap — no one stands on it, so it only needs to be solid, not pretty.)*

Step 6.5 — Skin, paint-prep, and levelers

1. Lay a ¼" Masonite square on the curved panel and trace the curved edge from below; cut the skin to match with the jigsaw.
2. Nail the skin down (as Step 5.5).
3. Add the 4 corner blocks + leveler feet (as Step 5.6). For the two back corners that were cut away, place the blocks/feet at the **nearest solid framing** inside the curve.

Each curved panel still has its cable tunnel and notch at the **front** edge — that's what keeps the tunnel continuous. ✅

Part 7 — Paint

1. Lightly sand every Masonite top and all exposed edges (120 grit). Wipe off dust.
2. Roll **two coats of matte black** (Behr Ultra Pure Black) on the tops and any visible edges. Let each coat dry per the can.
3. Paint the **MDF skirt pieces** (Part 9) black too — easier before they're mounted.

Paint the panels **before** final assembly so you don't get roller marks on assembled seams. Touch up seams after bolting.

Part 8 — Assemble on site

Now bolt the 15 panels together in place, in front of the LED wall.

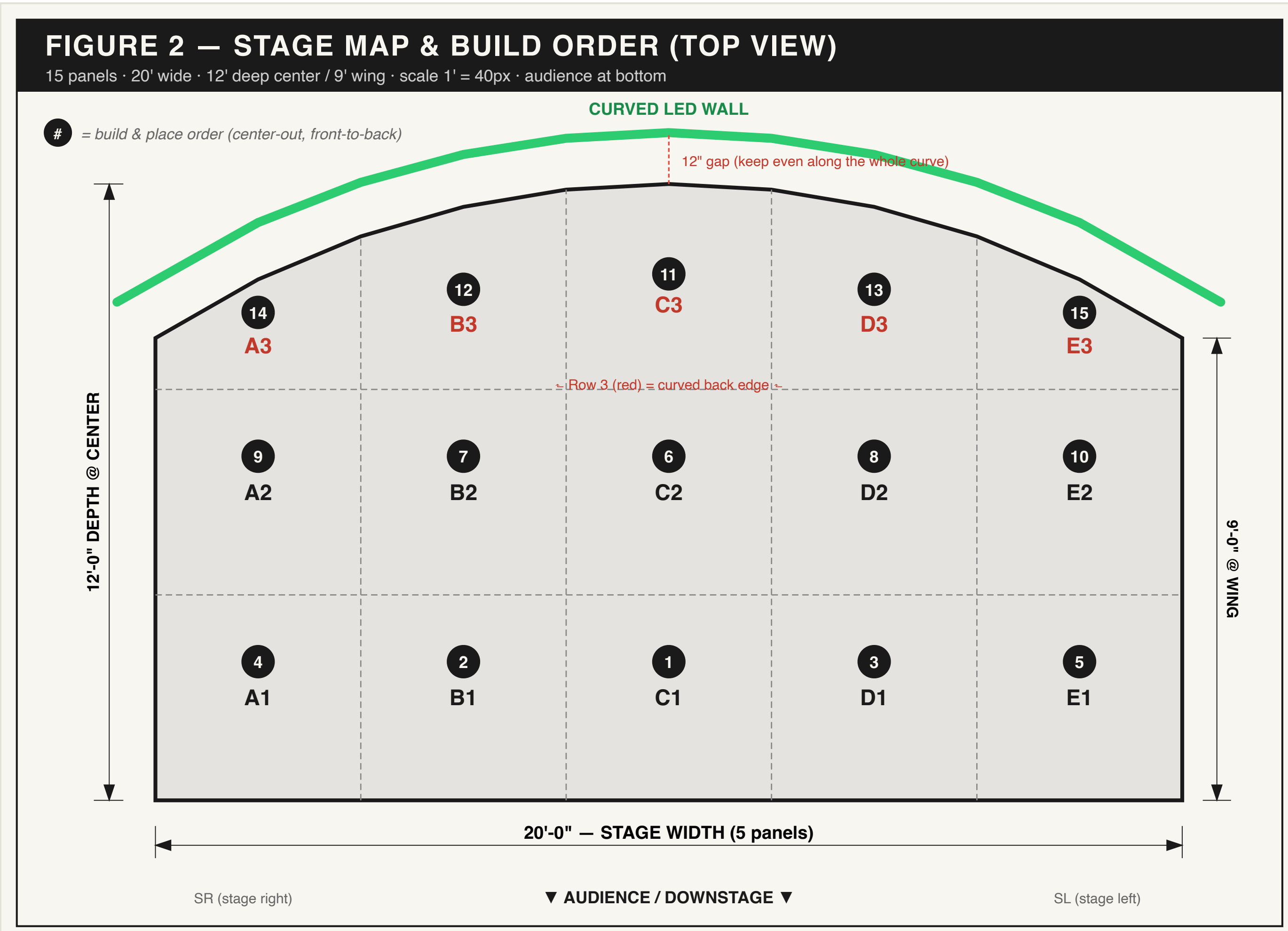

Figure 2 (again) — Placement order. Start at center-front (C1) and work outward and back.

Step 8.1 — Dry-place the grid

1. Set panels roughly in position in the order shown: **C1** → **B1** → **D1** → **A1** → **E1**, then row 2, then the curved row 3 last.
2. Leave the back (curved) edge **12" off the LED wall face** — measure the gap at several points along the curve. This 12" is for cables and sightlines; keep it consistent.

Step 8.2 — Align with dowel pins

1. Where two panels meet, drill **¼" holes ~2" deep** into the mating rail faces, **2 holes per seam** (about 6" in from each end), lined up across the seam.
2. Tap a **¼" × 2" dowel pin** halfway into one panel; the neighbor slips onto it. Dowels keep the tops perfectly flush at the seam.

Step 8.3 — Bolt panels together

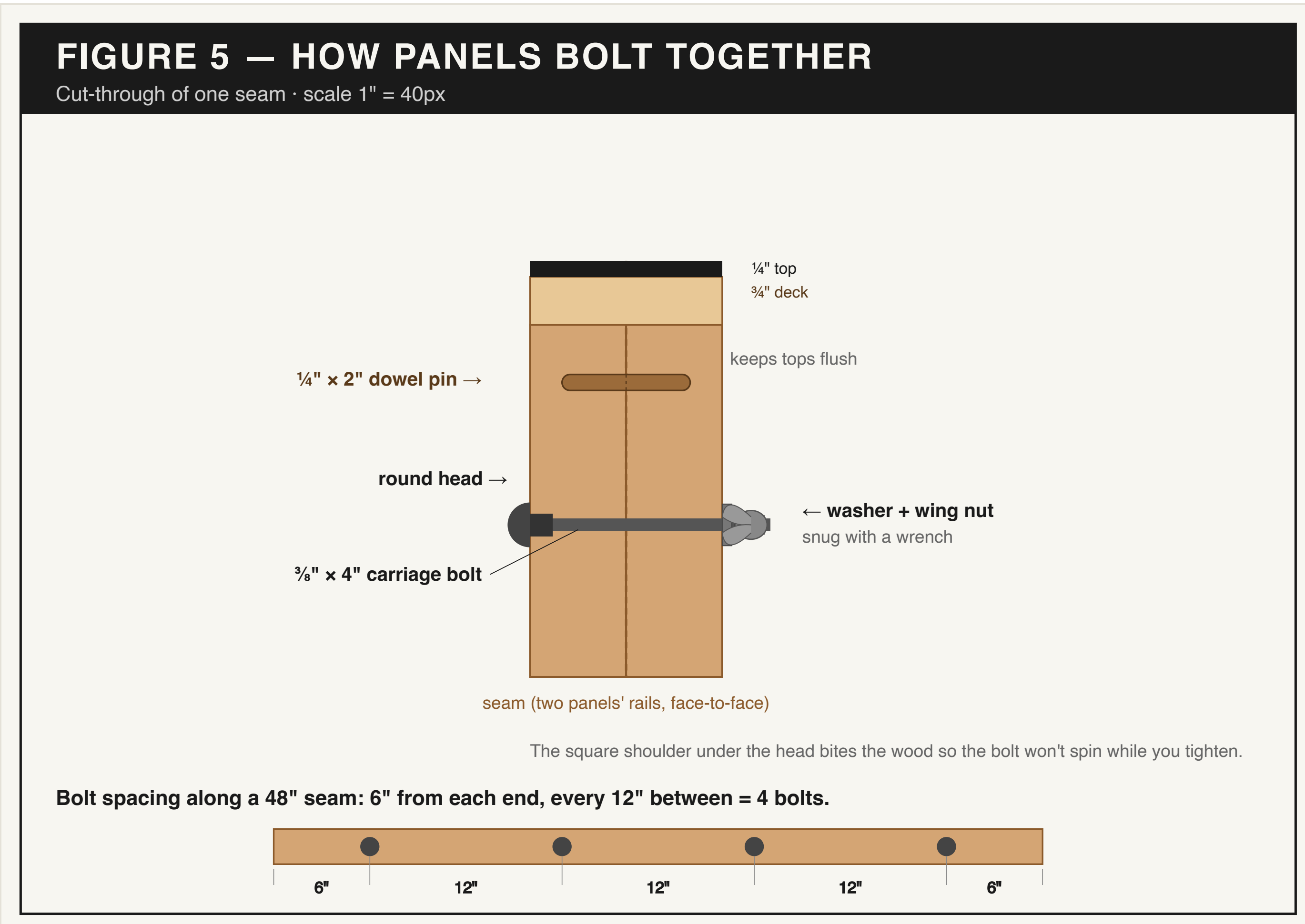

Figure 5 — A bolted seam: dowel pin for alignment, ⅜" carriage bolt + washer + wing nut to clamp.

1. Along each seam, drill **⅜" holes** through both mating rails, placed **6" from each end and every 12" between** (a full 48" seam gets **4 bolts**; shorter curved-row seams get 3).
2. Push a **⅜" × 4" carriage bolt** through, add a **washer + wing nut** on the far side, and tighten snug with the wrench. The square shoulder under the carriage-bolt head bites the wood so it won't spin.
3. **Avoid the tunnel:** keep bolts clear of the 6" center notch (the 18" and 30" positions sit just outside it — good).

Bolt count check: 5 front-row + 5 mid-row column seams, 8 between-row seams, plus curved-row seams ≈ **86 bolts** used. The plan's ~180 leaves a big spare margin.

Step 8.4 — Level the whole stage

1. With everything bolted, put a level on top across multiple seams.
2. Adjust the **leveler feet** up or down (¾" stud, ½"–1½" range) until the entire 20' top is flat and seams are flush. Start from the highest corner and work out.
3. Re-snug all wing nuts after leveling.

Part 9 — Front skirt + cable-access doors

The skirt is the black face the audience sees, with pop-out doors to reach the cable tunnels.

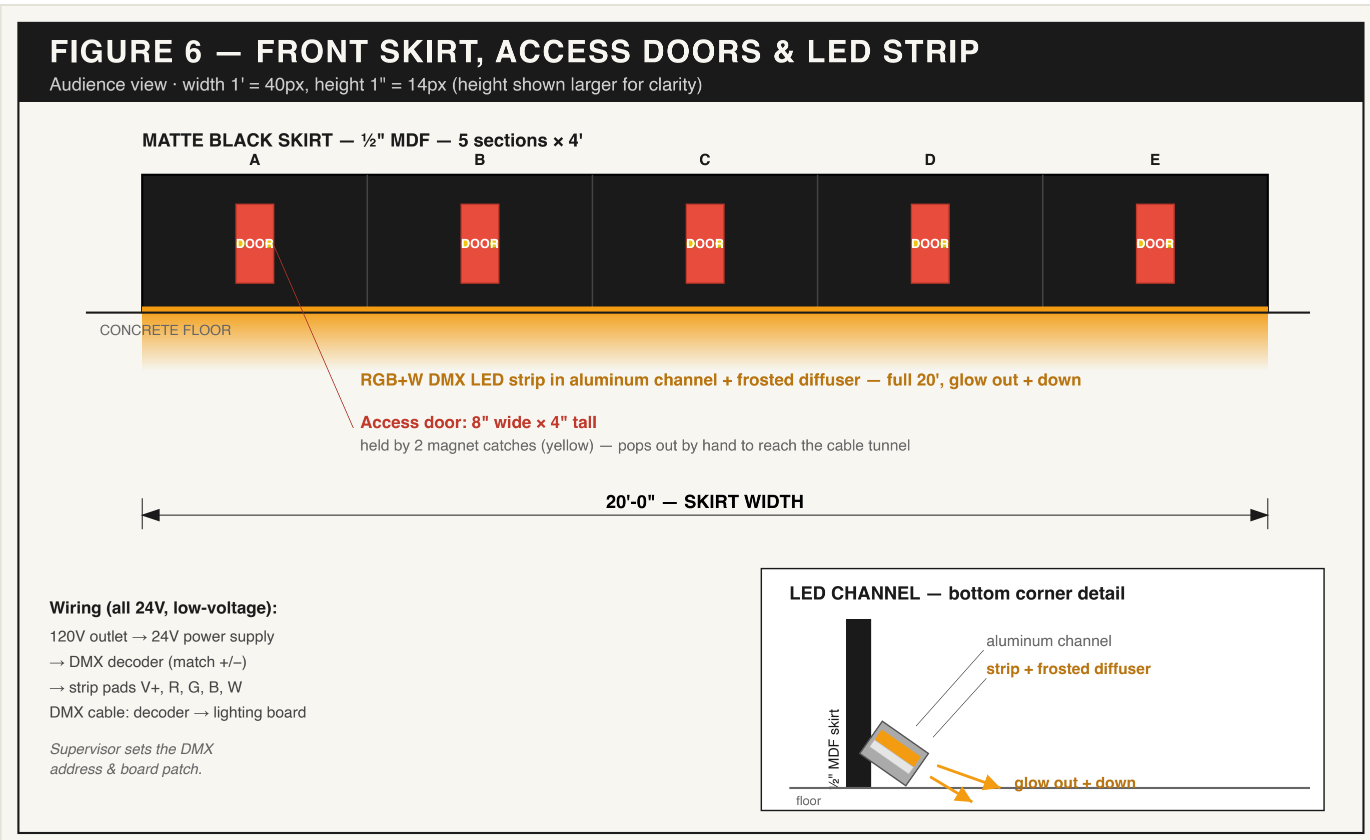

Figure 6 — Front skirt: 5 sections of ½" black MDF, each with a magnet-held access door over a cable tunnel, LED channel along the bottom.

1. You already cut **five 48" × 7"** skirt pieces from the ½" MDF (painted black).
2. In each skirt piece, cut a **8" wide × 4" tall** door opening, **centered**, lined up with that column's cable tunnel. Keep the cut-out piece — that's the door.
3. Mount each skirt section to the **downstage front rail** with **#8 × 1¼"** screws, every 12".
4. On each door, attach **2 neodymium magnet catches** (one each side) so it pops out by hand and snaps back. (10 catches ÷ 5 doors = 2 each.)

Part 10 — LED strip (downstage glow)

Low-voltage (24V) and safe to handle. If you're unsure about the DMX address, have your **Virtual Production / Unreal supervisor** set it (they run the lighting board).

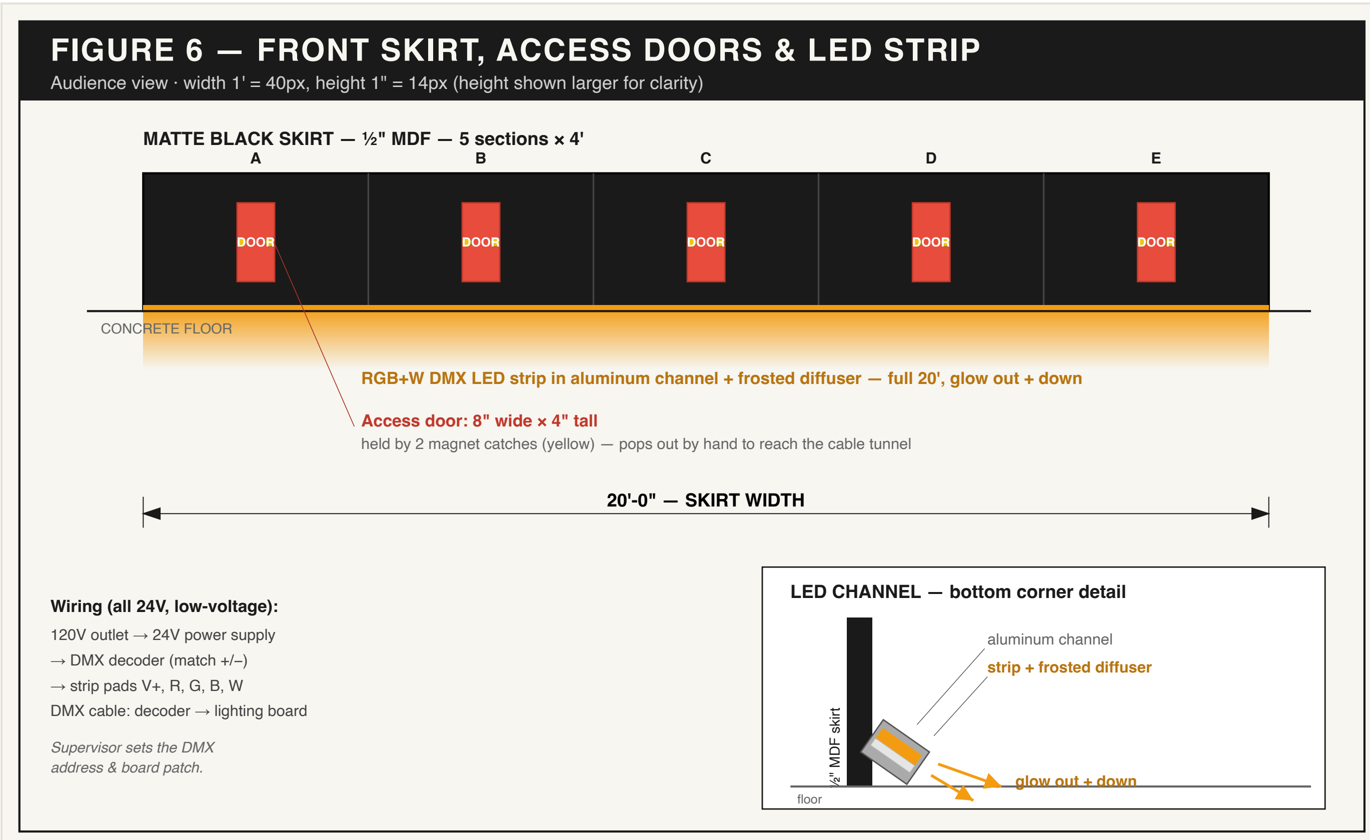

Figure 6 (again) — The aluminum LED channel mounts along the bottom of the front skirt, glow pointing out and down.

1. **Mount the channel:** screw the aluminum channel along the **bottom of the front skirt**, full 20', glow facing **out and slightly down**. Cut channel to length with a hacksaw; corners can butt.
2. **Lay the strip:** peel-and-stick the **RGB+W strip** into the channel. Keep it as one continuous run if you can; if you must cut, only cut on the marked cut-lines and rejoin with solderless connectors.
3. **Snap on the frosted diffuser.**
4. **Wire it** (all low-voltage): - **120V wall outlet** → **24V power supply** (just plug it in — no hardwiring). - **Power supply 24V+ / 24V–** → **DMX decoder** input (match + to +, – to –). - **DMX decoder outputs** → **strip pads** (V+, R, G, B, W — match the labels). - **5-pin DMX cable** from the decoder's DMX-in to a DMX-out on the lighting board. Run it through the cable tunnel.
5. **Set the DMX address** on the decoder and patch it on the board (supervisor confirms).
6. **Test** all colors + white before tucking the wiring away. Route the power supply and decoder into a tunnel bay, reachable through a skirt door.

> **Power check:** 7 m of 60-LED/m RGBW pulls roughly 130 W at full white — the **200 W** supply runs it at about ⅔ load. Good headroom. ✅

Part 11 — Final checks

- ☐ Walk the whole stage — no rocking panels, no high seams (re-level any that move).
- ☐ All wing nuts snug.
- ☐ Top is flat and clean; touch up paint on seams.
- ☐ Every skirt door pops out and snaps back; tunnels reachable.
- ☐ LED strip runs all colors + white from the board.
- ☐ **12" gap to the LED wall** is even along the whole curve.
- ☐ No screw heads proud of any walking surface.

Maintenance & knock-down

Because it's bolted (not glued), the stage comes apart:

- **To re-cable:** open a skirt door and reach the tunnel. No disassembly needed.
- **To remove a panel:** unbolt its seams (wing nuts), lift out. Two people.
- **Re-level** any time the floor or seams shift — just turn the feet.
- Keep a small bag of spare bolts, wing nuts, washers, and dowels (you have plenty left over) on top of the cable rack.

Appendix A — Verified measurements (my double-check)

For confidence, here's what was checked and confirmed against the blueprint:

Check	Result
Deck stack height	¼" + ¾" + 5½" + ½" leveler = **7.0"** ✅
Grid width	5 × 48" = 240" = **20'-0"** ✅
Depth at center	3 × 48" = 144" = **12'-0"** ✅
Curve radius (R)	**218"** — gives center 144" & wings 108" exactly ✅ (blueprint's 217.85" was rounding)
Depth formula	depth = $\sqrt{(47{,}524 - x^2)} - 170$, x = inches from center
Depth at wings	$\sqrt{(47{,}524 - 120^2)} - 74$ = **108" = 9'-0"** ✅
Cable tunnel	6" clear between tunnel rails (at 21"–27"); 6"×4" notch in rails ✅
Side bays	18" clear each — fine for ¾" ply deck span ✅
Neighbor-edge depths	B3↔C3 match at $46\frac{11}{16}$"; A3↔B3 match at 35¾" → curve is continuous ✅
LED power	7 m RGBW ≈ 130 W on a 200 W supply ✅

Appendix B — Fastener schedule (what goes where)

Fastener	Where	Roughly
#8 × 3" deck screws	Frame joints (3/joint) + deck-to-frame (every 8")	~1000
#8 × 1¼" screws	Skirt, leveler corner blocks, magnet catches	~500
#6 × ⅝" finish nails	Masonite top (6" edges / 12" field)	1 box
⅜" × 4" carriage bolt + wing nut + washer	Panel-to-panel seams (4 per 48" seam)	~90 used
¼" × 2" dowel pins	Seam alignment (2 per seam)	~44 used
Stage leveler feet	4 per panel	60
Magnet catches	2 per skirt door	10

Related files

- Stage blueprint — the design drawing this manual builds
- messenger-studios-stage-materials-quote — costed bill of materials (~$2,800)
- messenger-studios-led-wall — the LED wall the stage sits in front of
- messenger-studios-overview — full studio gear & systems reference